Esercizio 1. *The class of e.c. models of $\mathcal{T}$ admits amalgamation.*

Proof. Let $f : A \to B$ and $g : A \to C$ maps between e.c. models of $\mathcal{T}$. We want to find $D \models \mathcal{T}$ (not necessarily e.c. yet) such that there exist $h : B \to D$ and $l : C \to D$ witnessing amalgamation. We just need to prove that the following type is consistent:

$$\mathcal{T} \cup at - tp(B) \cup at - tp(C)$$

where elements of the form $f(a), g(a)$ are nominated as a. This can be done because (since A, B, C e.c.) f, g, h, l embeddings so we can identify an element with its image. Then taking $h(f(a)) = a^D = l(g(a))$ we will have amalgamation on D. Then consider the direct limit construction to have $i : D \to E$. This gives that $i \circ h$ and $i \circ l$ are the desired maps. Now, by contradiction assume the above type is inconsistent. Then it is finitely inconsistent. Let $\psi_1(b, a), .., \psi_n(b, a) \in at - tp(B)$ and $\theta(c, a), .., \theta(c, a) \in at - tp(C)$ with b, c tuple of constants different from those nominated by A, such that

$$\mathcal{T} \cup \{ \bigwedge_{i=1,..,n} \psi_i(b, a)\} \cup \{ \bigwedge_{j=1,..,m} \theta_j(c, a)\}$$

inconsistent. This is the same as

$$\mathcal{T} \cup \{\exists y \bigwedge_{i=1,..,n} \psi_i(y, a)\} \cup \{\exists z \bigwedge_{j=1,..,m} \theta_j(z, a)\}$$

inconsistent. But by hypothesis A is e.c. model of $\mathcal{T}$ and, as $B \models \exists y \bigwedge_{i=1,..,n} \psi(y, f(a))$ and $C \models \exists z \bigwedge_{j=1,..,m} \theta_j(z, g(a))$, we have that A is a model of the above type, contradiction.

□